Hamsika Rajeshwar Rao Garugula

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TECHNICAL SKILLS

Programming & Data: Python, TypeScript, JavaScript, SQL, PySpark, Databricks, NumPy, Pandas, PostgreSQL, MySQL, BigQuery, Snowflake, REST APIs
Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Gradient Boosting, FastAPI, Spring Boot, React, Node.js, Pydantic, Streamlit, Vite, Tailwind, SSE
Machine Learning & LLMs: Multi-Agent Orchestration, Human-in-the-Loop (HITL), Guardrails, Function/Tool Calling, Structured Outputs, Responses API, MCP (Model Context Protocol), LangChain, LangGraph, ReAct, RAG, RAGAS, Hybrid Search, Embeddings, Model Evaluation, Golden Datasets, LLM-as-Judge, Regression Suites, Hallucination Detection, Prompt Engineering, Fine-Tuning, Vector DBs (Qdrant, FAISS, ChromaDB, Weaviate), OpenAI API, Anthropic API, Gemini API, Transformers, Hugging Face, Reinforcement Learning (RL), Deep Learning
MLOps & Cloud: AWS, GCP, Azure, Weights & Biases, SageMaker, Airflow, Kafka, ML Infrastructure, Model Deployment, Model Monitoring
Developer Tools: Git, CI/CD, Docker, Kubernetes, Claude Code, Codex

EXPERIENCE

AI / Data Engineer | Saayam for All | February 2026 – Present

- Built an LLM routing layer with a zero-shot classifier (facebook/bart-large-mnli) that dispatches requests across 4 model providers by intent, shipping a model-powered feature end to end from intent classification through provider orchestration. Multi-model assistant supporting Meta AI, Gemini, ChatGPT, and Grok.
- Built a real-time evaluation and monitoring pipeline across 4 LLM providers, root-causing an **18x latency variance** (Grok 0.856s vs Meta AI 15.2s) and **30x throughput difference** (630 vs 20 tokens/sec) to improve latency, cost, and reliability of model serving.
- Built JavaScript frontend with real-time Markdown rendering via marked.js, live per-model performance metrics visualization.

AI & ML Engineer | L&T Technology Services | June 2023 – August 2023

- Engineered a production-grade OCR pipeline in Python with automated preprocessing across 18+ regional plate formats, achieving **85% format compatibility** and reducing error rates by **25%** at 300+ DPI.
- Optimized image preprocessing stack (normalization, skew correction, noise removal via OpenCV) improving OCR accuracy by **30%** across diverse regional formats.
- Performed training-data curation and quality evaluation, annotating and validating datasets for BMW ADAS computer-vision model training, improving object detection precision on safety-critical tasks.

Software Engineer | Kleza Solutions Pvt Ltd | July 2022 – March 2023

- Architected and shipped RESTful microservices for a healthcare Care Assessment App (Spring Boot + PostgreSQL + MongoDB), cutting API response times by **50%** through query optimization and service-layer redesign.
- Deployed a production ML system (XGBoost) for real-time clinical risk stratification into a live backend pipeline, enabling automated high-risk patient flagging and follow-up routing.

PROJECTS

Research-and-Report Agents | *Python · OpenAI Agents SDK · MCP · Responses API · Pydantic · FastAPI · Pytest*

- Built a three-agent system (Planner to Executor to Reviewer) on the **OpenAI Agents SDK** coordinating via handoffs with state management, retries, and a human-in-the-loop approval gate, exposing data and actions through a **self-built MCP server**, ending every run in a typed, guardrailed ReportResult.
- Built the evaluation layer the system is judged on: a **15-case golden dataset with an LLM-as-judge and a pytest regression suite** scoring accuracy, groundedness, and faithfulness at **0.95**, and after finding my own judge penalized correct-but-detailed answers as ungrounded, rewrote the rubric to flag only contradictions, raising eval from 0.80 to 0.95.

Multi-Agent Coding Assistant | *Python · FastAPI · LangGraph · Anthropic API · React · TypeScript · Tailwind · Vite · SSE*

- Built generative AI agents as a **5-agent LangGraph/FastAPI pipeline** using LLM orchestration frameworks (Planner to Architect to Coder to 2x Reviewer) that generates multi-file codebases from one prompt, shipping a **25-file React/TypeScript app** in a single run, with a **self-correcting review loop** (up to 3 retries) that caught and repaired broken output before it shipped.
- **Cut LLM inference cost** by routing per task: Opus for generation (~82% of per-run cost), Sonnet for planning/review (a full dual-reviewer cycle ran **$0.0007**), with a Haiku intent classifier fronting both code-gen and chat from one input.
- Streamed per-agent telemetry (tokens, latency, cost) over **SSE**, surfacing the real bottleneck: ~8.7s coder vs ~1.5s reviewer.

RAG Evaluation Harness | *Python · FastAPI · FAISS · Claude API · Sentence Transformers · Pydantic · Pytest*

- Built an **automated model evaluation framework** using an LLM-as-judge to replace manual scoring, **cutting evaluation effort ~90%**. On a 75-question benchmark: **faithfulness 0.996**, answer relevancy 0.82, **hallucination 0.017** (98 claims verified, 0.98 grounding).
- Built a pluggable multi-backend retrieval layer (FAISS, ChromaDB, Qdrant, Weaviate) behind a FastAPI gateway, making cross-backend benchmarking a one-config change. Benchmarked 75 queries at **Recall@5 0.87 / Recall@10 0.92**, **MRR 0.75, NDCG@10 0.79**.

RL-Driven Decision Making for Derivatives Markets | Python *· PyTorch · OpenAI Gym · Scikit-learn*

- Engineered a custom OpenAI Gym **reinforcement learning (RL) environment** with reward design, and built a **DDQN agent with prioritized experience replay** achieving **15% ROI** with zero major drawdowns and 800-step test stability on NVDA call option data.

EDUCATION

M.S. Artificial Intelligence	August 2024 – May 2026
University at Buffalo, SUNY	*Buffalo, New York*
B.E. Electronics & Communication Engineering	December 2020 – May 2024
Chaitanya Bharathi Institute of Technology	*Hyderabad, India*

PUBLICATIONS

Survey on Antennas for Different Cancers and Tumor Detection, ISDA 2023, Springer. doi.org/10.1007/978-3-031-64776-5_31